

SAFETY DATA SHEET

TALSTAR 250 EC



Version	Revision Date:	SDS Number:	Date of last issue: 22.12.2021
1.1	17.12.2025	50001563	Date of first issue: 22.12.2021

SECTION 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : TALSTAR 250 EC

Other means of identification : BIFENTHRIN 250 G/L EC

Recommended use of the chemical and restrictions on use

Recommended use : Can be used as insecticide only.

Restrictions on use : Use as recommended by the label.

Manufacturer or supplier's details

Company : FMC Australasia Pty Ltd

Address : Building B, Suite G.01, 12 Julius Avenue
North Ryde NSW 2113
Australia

Telephone : 1 800 066 355

Telefax : (02) 9923 6011

E-mail address : SDS-Info@fmc.com

Emergency telephone number : For leak, fire, spill or accident emergencies, call:
1800 033 111 (Ixm)

Medical emergency:
1 800 033 111 (Transport and 24 h Medical information)

SECTION 2. HAZARDS IDENTIFICATION

GHS Classification

Flammable liquids : Category 4

Acute toxicity (Oral) : Category 3

Acute toxicity (Inhalation) : Category 4

Carcinogenicity : Category 2

Specific target organ toxicity - single exposure : Category 1 (Central nervous system)

SAFETY DATA SHEET

TALSTAR 250 EC



Version	Revision Date:	SDS Number:	Date of last issue:
1.1	17.12.2025	50001563	22.12.2021
			Date of first issue: 22.12.2021

Specific target organ toxicity - single exposure : Category 3 (Central nervous system)

Specific target organ toxicity - repeated exposure : Category 1 (Central nervous system)

Aspiration hazard : Category 1

GHS label elements

Hazard pictograms :  

Signal word : Danger

Hazard statements : H227 Combustible liquid.
H301 Toxic if swallowed.
H304 May be fatal if swallowed and enters airways.
H332 Harmful if inhaled.
H336 May cause drowsiness or dizziness.
H351 Suspected of causing cancer.
H370 Causes damage to organs (Central nervous system).
H372 Causes damage to organs (Central nervous system) through prolonged or repeated exposure.

Precautionary statements : **Prevention:**
P201 Obtain special instructions before use.
P202 Do not handle until all safety precautions have been read and understood.
P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P260 Do not breathe mist or vapours.
P264 Wash skin thoroughly after handling.
P270 Do not eat, drink or smoke when using this product.
P271 Use only outdoors or in a well-ventilated area.
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection/ hearing protection.

Response:

P301 + P310 + P330 IF SWALLOWED: Immediately call a POISON CENTER/ doctor. Rinse mouth.
P304 + P340 + P312 IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER/ doctor if you feel unwell.
P308 + P311 IF exposed or concerned: Call a POISON CENTER/ doctor.
P331 Do NOT induce vomiting.
P370 + P378 In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.

Storage:

P403 + P233 Store in a well-ventilated place. Keep container

SAFETY DATA SHEET

TALSTAR 250 EC



Version 1.1 Revision Date: 17.12.2025 SDS Number: 50001563 Date of last issue: 22.12.2021
Date of first issue: 22.12.2021

tightly closed.
P405 Store locked up.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification

Very toxic to aquatic life with long lasting effects.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Chemical name	CAS-No.	Concentration (% w/w)
bifenthrin (ISO)	82657-04-3	≥ 10 -< 30
Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified	64742-94-5	≥ 60 -<= 100
calcium dodecylbenzenesulphonate	26264-06-2	≥ 3 -< 10
2-ethylhexan-1-ol	104-76-7	< 10

SECTION 4. FIRST AID MEASURES

- General advice : Move out of dangerous area.
Consult a physician.
Show this safety data sheet to the doctor in attendance.
Symptoms of poisoning may appear several hours later.
Do not leave the victim unattended.
- If inhaled : Consult a physician after significant exposure.
If unconscious, place in recovery position and seek medical advice.
- In case of skin contact : If skin irritation persists, call a physician.
If on skin, rinse well with water.
If on clothes, remove clothes.
- In case of eye contact : Flush eyes with water as a precaution.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.
- If swallowed : Keep respiratory tract clear.
Do NOT induce vomiting.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.
Take victim immediately to hospital.

Most important symptoms : Toxic if swallowed.

SAFETY DATA SHEET



TALSTAR 250 EC

Version 1.1	Revision Date: 17.12.2025	SDS Number: 50001563	Date of last issue: 22.12.2021 Date of first issue: 22.12.2021
----------------	------------------------------	-------------------------	---

and effects, both acute and delayed

May be fatal if swallowed and enters airways.
Harmful if inhaled.
May cause drowsiness or dizziness.
Suspected of causing cancer.
Causes damage to organs.
Causes damage to organs through prolonged or repeated exposure.

Notes to physician : Treat symptomatically.

SECTION 5. FIREFIGHTING MEASURES

Suitable extinguishing media : Carbon dioxide (CO₂)
Foam
Dry chemical

Unsuitable extinguishing media : High volume water jet

Specific hazards during fire-fighting : Do not allow run-off from fire fighting to enter drains or water courses.

Hazardous combustion products : Fire may produce irritating, corrosive and/or toxic gases.
Carbon oxides
Fluorinated compounds
Chlorinated compounds
Hydrogen chloride
Hydrogen fluoride
Nitrogen oxides (NO_x)
Sulphur oxides

Specific extinguishing methods : Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.
For safety reasons in case of fire, cans should be stored separately in closed containments.
Use a water spray to cool fully closed containers.

Special protective equipment for firefighters : Wear self-contained breathing apparatus for firefighting if necessary.

Hazchem Code : 2X

SECTION 6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures : Use personal protective equipment.
Ensure adequate ventilation.

Environmental precautions : Prevent product from entering drains.
Prevent further leakage or spillage if safe to do so.
If the product contaminates rivers and lakes or drains inform

TALSTAR 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: 22.12.2021
1.1	17.12.2025	50001563	Date of first issue: 22.12.2021

respective authorities.

Methods and materials for containment and cleaning up : Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13).
Keep in suitable, closed containers for disposal.

SECTION 7. HANDLING AND STORAGE

Advice on protection against fire and explosion : Do not spray on a naked flame or any incandescent material.
Keep away from open flames, hot surfaces and sources of ignition.

Advice on safe handling : Avoid formation of aerosol.
Do not breathe vapours/dust.
Avoid exposure - obtain special instructions before use.
Avoid contact with skin and eyes.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Provide sufficient air exchange and/or exhaust in work rooms.
Dispose of rinse water in accordance with local and national regulations.

Hygiene measures : Avoid contact with skin, eyes and clothing.
When using do not eat or drink.
When using do not smoke.
Wash hands before breaks and immediately after handling the product.

Conditions for safe storage : Prevent unauthorized access.
No smoking.
Keep container tightly closed in a dry and well-ventilated place.
Containers which are opened must be carefully resealed and kept upright to prevent leakage.
Observe label precautions.
Electrical installations / working materials must comply with the technological safety standards.

Further information on storage stability : No decomposition if stored and applied as directed.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION**Components with workplace control parameters**

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis
Solvent naphtha (petroleum),	64742-94-5	TWA	200 mg/m3	ACGIH

SAFETY DATA SHEET

TALSTAR 250 EC



Version 1.1	Revision Date: 17.12.2025	SDS Number: 50001563	Date of last issue: 22.12.2021 Date of first issue: 22.12.2021
----------------	------------------------------	-------------------------	---

heavy arom.; Kerosine — un-specified			(total hydrocarbon vapor)	
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Personal protective equipment

Respiratory protection : In case of mist, spray or aerosol exposure wear suitable personal respiratory protection and protective suit.

Hand protection

Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Eye protection : Eye wash bottle with pure water
Tightly fitting safety goggles

Skin and body protection : Impervious clothing
Choose body protection according to the amount and concentration of the dangerous substance at the work place.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state : liquid

Form : liquid

Colour : light yellow

Odour : Aromatic

pH : 5.5 - 7.5

Melting point/freezing point : No data available

Boiling point/boiling range : No data available

Flash point : No data available

Self-ignition : No data available

Density : 0.97 g/cm³

Solubility(ies)
Water solubility : emulsifiable

Viscosity
Viscosity, kinematic : No data available

SAFETY DATA SHEET

TALSTAR 250 EC



Version	Revision Date:	SDS Number:	Date of last issue: 22.12.2021
1.1	17.12.2025	50001563	Date of first issue: 22.12.2021

Explosive properties : Not explosive

Oxidizing properties : Non-oxidizing

SECTION 10. STABILITY AND REACTIVITY

Reactivity : No decomposition if stored and applied as directed.

Chemical stability : No decomposition if stored and applied as directed.

Possibility of hazardous reactions : No decomposition if stored and applied as directed.
Vapours may form explosive mixture with air.

Conditions to avoid : Heat, flames and sparks.

Incompatible materials : Strong acids
Strong oxidizing agents
Strong bases

Hazardous decomposition products : No hazardous decomposition products are known.

SECTION 11. TOXICOLOGICAL INFORMATION

Acute toxicity

Toxic if swallowed.

Harmful if inhaled.

Product:

Acute oral toxicity : LD50 (Rat): 250 mg/kg
Remarks: Based on data from similar materials

Acute inhalation toxicity : LC50 (Rat): 1.6 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Remarks: Based on data from similar materials

Acute dermal toxicity : LD50 (Rat): > 2,000 mg/kg
Remarks: Based on data from similar materials

Components:

bifenthrin (ISO):

Acute oral toxicity : LD50 (Rat, male and female): 56.7 mg/kg
Symptoms: Convulsions, Tremors, ataxia

LD50 (Mouse, female): 42.5 mg/kg
Method: OPPTS 870.1100

TALSTAR 250 EC

Version	Revision Date:	SDS Number:	Date of last issue:
1.1	17.12.2025	50001563	22.12.2021
			Date of first issue: 22.12.2021

Acute inhalation toxicity : LC50 (Rat, female): 0.6 - 1.2 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
Symptoms: Tremors, Convulsions

LC50 (Rat, male): 1.10 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
Symptoms: Tremors, Fatality

Acute dermal toxicity : LD50 (Rat, male and female): > 2,000 mg/kg
Remarks: no mortality

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Acute oral toxicity : LD50 (Rat, male and female): > 5,000 mg/kg
Method: OECD Test Guideline 401
Remarks: Based on data from similar materials

Acute inhalation toxicity : LC50 (Rat, male and female): > 5.28 mg/l
Exposure time: 4 h
Test atmosphere: vapour
Assessment: The substance or mixture has no acute inhalation toxicity
Remarks: Based on data from similar materials

Acute dermal toxicity : LD50 (Rabbit, male and female): > 2,000 mg/kg
Method: OECD Test Guideline 402
Assessment: The substance or mixture has no acute dermal toxicity
Remarks: Based on data from similar materials

calcium dodecylbenzenesulphonate:

Acute oral toxicity : LD50 (Rat, male and female): 1,300 mg/kg
Remarks: Based on data from similar materials

Acute inhalation toxicity : Remarks: Not classified

Acute dermal toxicity : LD50 (Rat, male and female): > 2000 milligram per kilogram
Method: OECD Test Guideline 402
Assessment: The substance or mixture has no acute dermal toxicity
Remarks: Based on data from similar materials

2-ethylhexan-1-ol:

Acute oral toxicity : LD50 (Rat, male): 2,047 mg/kg

Acute inhalation toxicity : LC50 (Rat): 4.3 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist

SAFETY DATA SHEET



TALSTAR 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: 22.12.2021
1.1	17.12.2025	50001563	Date of first issue: 22.12.2021

Acute dermal toxicity : LD50 (Rat, male and female): > 3,000 mg/kg
Method: OECD Test Guideline 402
Assessment: The substance or mixture has no acute dermal toxicity

Skin corrosion/irritation

Based on available data, the classification criteria are not met.

Product:

Species : Rabbit
Result : No skin irritation
Remarks : Based on data from similar materials

Components:

bifenthrin (ISO):

Species : Rabbit
Method : OECD Test Guideline 404
Result : slight or no skin irritation.
GLP : yes

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Species : Rabbit
Result : No skin irritation

calcium dodecylbenzenesulphonate:

Species : Rabbit
Method : OECD Test Guideline 404
Result : Skin irritation

2-ethylhexan-1-ol:

Species : Rabbit
Method : OECD Test Guideline 404
Result : Skin irritation

Serious eye damage/eye irritation

Based on available data, the classification criteria are not met.

Product:

Species : Rabbit
Result : No eye irritation
Remarks : Based on data from similar materials

Components:

bifenthrin (ISO):

Species : Rabbit
Result : Slight or no eye irritation
Method : OECD Test Guideline 405

SAFETY DATA SHEET

TALSTAR 250 EC



Version	Revision Date:	SDS Number:	Date of last issue: 22.12.2021
1.1	17.12.2025	50001563	Date of first issue: 22.12.2021

GLP : yes

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Species : Rabbit
Result : No eye irritation
Remarks : Based on data from similar materials

calcium dodecylbenzenesulphonate:

Species : Rabbit
Result : Irreversible effects on the eye
Method : OECD Test Guideline 405
Remarks : Based on data from similar materials

Species : Rabbit
Result : Irreversible effects on the eye
Method : OECD Test Guideline 405

2-ethylhexan-1-ol:

Species : Rabbit
Result : Irritation to eyes, reversing within 21 days
Method : OECD Test Guideline 405

Respiratory or skin sensitisation

Skin sensitisation

Based on available data, the classification criteria are not met.

Respiratory sensitisation

Based on available data, the classification criteria are not met.

Product:

Species : Guinea pig
Result : Not a skin sensitizer.
Remarks : Based on data from similar materials

Components:

bifenthrin (ISO):

Test Type : Maximisation Test
Exposure routes : Skin contact
Species : Guinea pig
Method : OECD Test Guideline 406
Result : May cause sensitisation by skin contact.
GLP : yes

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Test Type : Buehler Test
Species : Guinea pig
Result : Does not cause skin sensitisation.
Remarks : Based on data from similar materials

TALSTAR 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: 22.12.2021
1.1	17.12.2025	50001563	Date of first issue: 22.12.2021

calcium dodecylbenzenesulphonate:

Test Type	: Maximisation Test
Species	: Guinea pig
Method	: OECD Test Guideline 406
Result	: Not a skin sensitizer.
Remarks	: Based on data from similar materials

Chronic toxicity**Germ cell mutagenicity**

Based on available data, the classification criteria are not met.

Components:**bifenthrin (ISO):**

Genotoxicity in vitro	: Test Type: Ames test Test system: Salmonella typhimurium Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 471 Result: negative Test Type: gene mutation test Test system: Chinese hamster ovary cells Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 476 Result: negative GLP: yes Test Type: Mouse lymphoma assay Metabolic activation: with and without metabolic activation Result: negative
Genotoxicity in vivo	: Test Type: Sex-linked Recessive Lethal Test Species: Drosophila melanogaster (vinegar fly) Result: negative Test Type: unscheduled DNA synthesis assay Species: Rat Method: OECD Test Guideline 486 Result: negative Test Type: Mammalian erythrocyte micronucleus test (in vivo cytogenetic assay) Species: Mouse (male and female) Method: OECD Test Guideline 474 Result: negative

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Genotoxicity in vitro	: Test Type: reverse mutation assay Result: negative Remarks: Based on data from similar materials
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SAFETY DATA SHEET



TALSTAR 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: 22.12.2021
1.1	17.12.2025	50001563	Date of first issue: 22.12.2021

Genotoxicity in vivo : Test Type: sister chromatid exchange assay
Species: Mouse
Application Route: Intraperitoneal injection
Result: negative
Remarks: Based on data from similar materials

calcium dodecylbenzenesulphonate:

Genotoxicity in vitro : Test Type: reverse mutation assay
Method: OECD Test Guideline 471
Result: negative
Remarks: Based on data from similar materials

Genotoxicity in vivo : Test Type: chromosome aberration assay
Species: Rat (male and female)
Application Route: Oral
Exposure time: 90 d
Result: negative
Remarks: Based on data from similar materials

Germ cell mutagenicity - Assessment : Weight of evidence does not support classification as a germ cell mutagen.

2-ethylhexan-1-ol:

Genotoxicity in vitro : Test Type: reverse mutation assay
Method: OECD Test Guideline 471
Result: negative

Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse
Application Route: Intraperitoneal injection
Result: negative

Carcinogenicity

Suspected of causing cancer.

Components:

bifenthrin (ISO):

Species : Rat, female
Application Route : Oral
Exposure time : 2 Years
NOAEL : 3 mg/kg bw/day
Result : negative

Species : Mouse, male
Application Route : Oral
Exposure time : 18 month(s)
NOAEL : 7.6 mg/kg bw/day
Result : positive
Symptoms : malignant tumors

SAFETY DATA SHEET



TALSTAR 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: 22.12.2021
1.1	17.12.2025	50001563	Date of first issue: 22.12.2021

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Species	: Mouse
Application Route	: Dermal
Exposure time	: 104 weeks
Result	: negative
Remarks	: Based on data from similar materials

Carcinogenicity - Assessment	: Limited evidence of carcinogenicity in animal studies
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calcium dodecylbenzenesulphonate:

Species	: Rat, male and female
Application Route	: Oral
Exposure time	: 720 d
NOAEL	: 250 mg/kg body weight
Result	: negative
Remarks	: Based on data from similar materials

Carcinogenicity - Assessment	: Weight of evidence does not support classification as a carcinogen
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2-ethylhexan-1-ol:

Species	: Rat
Application Route	: Oral
Exposure time	: 24 month(s)
Result	: negative

Reproductive toxicity

Based on available data, the classification criteria are not met.

Components:

bifenthrin (ISO):

Effects on fertility	: Test Type: Two-generation study Species: Rat Application Route: Oral General Toxicity - Parent: NOAEL: 3 mg/kg bw/day General Toxicity F1: NOAEL: 5 mg/kg bw/day Result: negative
Effects on foetal development	: Test Type: Embryo-foetal development Species: Rabbit Application Route: Oral General Toxicity Maternal: NOAEL: 2.7 mg/kg bw/day Teratogenicity: NOAEL: 2.7 mg/kg bw/day Symptoms: Maternal effects Result: No teratogenic effects Test Type: Embryo-foetal development Species: Rat Application Route: Oral General Toxicity Maternal: NOAEL: 1 mg/kg bw/day

TALSTAR 250 EC

Version	Revision Date:	SDS Number:	Date of last issue:
1.1	17.12.2025	50001563	22.12.2021
			Date of first issue: 22.12.2021

Teratogenicity: NOAEL: 2 mg/kg bw/day

Result: No teratogenic effects

Species: Rat

Application Route: Oral

General Toxicity Maternal: LOAEL: 7.2 mg/kg bw/day

Developmental Toxicity: LOAEL: 7.2 mg/kg bw/day

Embryo-foetal toxicity: NOEL: 9.0 mg/kg bw/day

Method: OECD Test Guideline 426

Result: Animal testing did not show any effects on fertility.,
Some evidence of adverse effects on development, based on
animal experiments.

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Effects on fertility : Test Type: Fertility
Species: Rat, male and female
Application Route: Oral
Method: OECD Test Guideline 415
Result: negative
Remarks: Based on data from similar materials

Effects on foetal development : Test Type: reproductive and developmental toxicity study
Species: Rat
Application Route: Oral
Method: OECD Test Guideline 414
Result: negative
Remarks: Based on data from similar materials

calcium dodecylbenzenesulphonate:

Effects on fertility : Test Type: Fertility/early embryonic development
Species: Rat, male and female
Application Route: Ingestion
General Toxicity - Parent: NOAEL: 400 mg/kg body weight
Method: OECD Test Guideline 422
Result: negative

Effects on foetal development : Test Type: reproductive and developmental toxicity study
Species: Rat
Application Route: Ingestion
General Toxicity Maternal: NOAEL: 300 mg/kg body weight
Developmental Toxicity: NOAEL: 600 mg/kg body weight
Method: OECD Test Guideline 422
Result: negative

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

2-ethylhexan-1-ol:

Effects on foetal development : Test Type: Embryo-foetal development
Species: Mouse
Application Route: Oral
Method: OECD Test Guideline 414

SAFETY DATA SHEET

TALSTAR 250 EC



Version	Revision Date:	SDS Number:	Date of last issue: 22.12.2021
1.1	17.12.2025	50001563	Date of first issue: 22.12.2021

Result: negative

STOT - single exposure

May cause drowsiness or dizziness.
Causes damage to organs (Central nervous system).

Components:

bifenthrin (ISO):

Target Organs	: Central nervous system
Assessment	: Causes damage to organs.

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Assessment	: May cause drowsiness or dizziness.
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2-ethylhexan-1-ol:

Assessment	: May cause respiratory irritation.
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STOT - repeated exposure

Causes damage to organs (Central nervous system) through prolonged or repeated exposure.

Components:

bifenthrin (ISO):

Target Organs	: Central nervous system
Assessment	: The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 1.

Repeated dose toxicity

Components:

bifenthrin (ISO):

Species	: Rat, male and female
NOEL	: 100 ppm
Application Route	: Oral - feed
Exposure time	: 90 d
Remarks	: No toxicologically significant effects were found.

Species	: Dog, male and female
NOEL	: 2.5 mg/kg bw/day
Application Route	: Oral - feed
Exposure time	: 13 w
Symptoms	: Tremors

Species	: Dog, male and female
NOEL	: 1.5 mg/kg
Application Route	: Oral
Exposure time	: 52 w
Dose	: 0.75, 1.5, 3, 5 mg/kg bw/day
GLP	: yes
Symptoms	: Tremors

TALSTAR 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: 22.12.2021
1.1	17.12.2025	50001563	Date of first issue: 22.12.2021

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Species	: Rat, male and female
NOAEL	: 750 mg/kg
Application Route	: Oral - gavage
Exposure time	: 90 day
Remarks	: Based on data from similar materials

Species	: Rat, male and female
NOAEL	: 1 mg/l
LOAEL	: 0.5 mg/l
Application Route	: inhalation (vapour)
Exposure time	: 90 day
Symptoms	: Alpha-2u-globulin nephropathy

calcium dodecylbenzenesulphonate:

Species	: Rat, male and female
NOAEL	: 85 mg/kg
LOAEL	: 145 mg/kg
Application Route	: Oral
Exposure time	: 9 Months
Remarks	: Based on data from similar materials

Species	: Rat, male
LOAEL	: 286 mg/kg
Application Route	: Skin contact
Exposure time	: 15 Days
Remarks	: Based on data from similar materials

Species	: Rat, male and female
NOAEL	: 100 mg/kg bw/day
LOAEL	: 200 mg/kg bw/day
Application Route	: Oral - gavage
Exposure time	: 28 - 54 Days
Method	: OECD Test Guideline 422
Remarks	: Based on data from similar materials

2-ethylhexan-1-ol:

Species	: Rat
	: 250 mg/kg
Application Route	: Oral
Exposure time	: 13 Weeks
Method	: OECD Test Guideline 408

Aspiration toxicity

May be fatal if swallowed and enters airways.

Components:**bifenthrin (ISO):**

The substance does not have properties associated with aspiration hazard potential.

SAFETY DATA SHEET



TALSTAR 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: 22.12.2021
1.1	17.12.2025	50001563	Date of first issue: 22.12.2021

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

The substance or mixture is known to cause human aspiration toxicity hazards or has to be regarded as if it causes a human aspiration toxicity hazard.

Further information

Product:

Remarks : Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.
Concentrations substantially above the TLV value may cause narcotic effects.
Solvents may degrease the skin.

SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity

Components:

bifenthrin (ISO):

Toxicity to fish	: LC50 (Salmo gairdneri): 0.00015 mg/l Exposure time: 96 h Test Type: flow-through test LC50 (Lepomis macrochirus (Bluegill sunfish)): 0.00035 mg/l Exposure time: 96 h Test Type: flow-through test LC50 (Oncorhynchus mykiss (rainbow trout)): 0.000256 mg/l Exposure time: 96 h Test Type: semi-static test Method: OECD Test Guideline 203 GLP: yes LC50 (Pimephales promelas (fathead minnow)): 0.000234 mg/l Exposure time: 96 h Test Type: semi-static test Method: OECD Test Guideline 203 GLP: yes
Toxicity to daphnia and other aquatic invertebrates	: EC50 (Daphnia (water flea)): 0.00011 mg/l Exposure time: 48 h LC50 (Daphnia (water flea)): 0.0016 mg/l Exposure time: 48 h
Toxicity to algae/aquatic plants	: EC50 (algae): 0.822 mg/l Exposure time: 72 h
Toxicity to fish (Chronic tox-	: NOEC (Oncorhynchus mykiss (rainbow trout)): 0.00012 mg/l

SAFETY DATA SHEET

TALSTAR 250 EC



Version	Revision Date:	SDS Number:	Date of last issue:
1.1	17.12.2025	50001563	22.12.2021
			Date of first issue: 22.12.2021

icity) Exposure time: 21 d

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC (Daphnia magna (Water flea)): 0.0013 µg/l
Exposure time: 21 d

NOEC (Daphnia magna (Water flea)): 0.00095 µg/l
Exposure time: 21 d

Toxicity to soil dwelling organisms : LD50 (Eisenia fetida (earthworms)): > 16 mg/kg
Exposure time: 14 d

Toxicity to terrestrial organisms : LD50 (Colinus virginianus (Bobwhite quail)): 1,800 mg/kg

LD50 (Anas platyrhynchos (Mallard duck)): > 2,150 mg/kg

LD50 (Apis mellifera (bees)): 0.07 µg a.i./bee
Exposure time: 48 h
End point: Acute oral toxicity
Method: OECD Test Guideline 213

LD50 (Apis mellifera (bees)): 0.0201 µg a.i./bee
Exposure time: 48 h
End point: Acute contact toxicity
Method: OECD Test Guideline 214

ED50 (Apis mellifera (bees)): 143 ng ai larva
Exposure time: 7 d
End point: Acute larval toxicity
Method: OECD 237

ED50 (Apis mellifera (bees)): 15.15 ng ai larva/day
Exposure time: 22 d
End point: Chronic larval toxicity
Method: OECD 239

NOAEL (Apis mellifera (bees)): 2.30 ng ai larva/day
Exposure time: 22 d
End point: Chronic larval toxicity
Method: OECD 239

ED50 (Apis mellifera (bees)): 300 ng a.i./bee/day
Exposure time: 10 d
End point: Chronic oral toxicity
Method: OECD TG 245

NOAEL (Apis mellifera (bees)): 170 ng a.i./bee/day
Exposure time: 10 d
End point: Chronic oral toxicity
Method: OECD TG 245

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

SAFETY DATA SHEET

TALSTAR 250 EC



Version	Revision Date:	SDS Number:	Date of last issue:
1.1	17.12.2025	50001563	22.12.2021
			Date of first issue: 22.12.2021

- Toxicity to fish : LL50 (*Oncorhynchus mykiss* (rainbow trout)): 2 - 5 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203
Remarks: water accommodated fractions (WAF)
- Toxicity to daphnia and other aquatic invertebrates : EL50 (*Daphnia magna* (Water flea)): 1.4 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202
Remarks: water accommodated fractions (WAF)
- Toxicity to algae/aquatic plants : EL50 (*Pseudokirchneriella subcapitata* (green algae)): > 1 - 3 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201
Remarks: water accommodated fractions (WAF)
- Toxicity to microorganisms : LL50 (*Tetrahymena pyriformis*): 677.9 mg/l
Exposure time: 72 h
Test Type: Growth inhibition

calcium dodecylbenzenesulphonate:

- Toxicity to fish : LC50 (*Danio rerio* (zebra fish)): 10 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203
Remarks: Based on data from similar materials
- LC50 (*Pimephales promelas* (fathead minnow)): 4.6 mg/l
Exposure time: 96 h
Remarks: Based on data from similar materials
- Toxicity to daphnia and other aquatic invertebrates : EC50 (*Daphnia magna* (Water flea)): 3.5 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202
Remarks: Based on data from similar materials
- Toxicity to algae/aquatic plants : NOEC (*Pseudokirchneriella subcapitata* (green algae)): 7.9 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201
Remarks: Based on data from similar materials
- EC50 (*Pseudokirchneriella subcapitata* (green algae)): 65.4 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201
Remarks: Based on data from similar materials
- Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC (*Daphnia magna* (Water flea)): 1.65 mg/l
Exposure time: 21 d
Remarks: Based on data from similar materials
- NOEC (*Daphnia magna* (Water flea)): 1.18 mg/l
Exposure time: 21 d

SAFETY DATA SHEET

TALSTAR 250 EC



Version	Revision Date:	SDS Number:	Date of last issue:
1.1	17.12.2025	50001563	22.12.2021
			Date of first issue: 22.12.2021

Remarks: Based on data from similar materials

- Toxicity to microorganisms : EC50 (activated sludge): 500 mg/l
Exposure time: 3 h
Method: OECD Test Guideline 209
- Toxicity to soil dwelling organisms : LC50 (*Eisenia fetida* (earthworms)): 1,000 mg/kg
Exposure time: 14 d
Method: OECD Test Guideline 207
- Toxicity to terrestrial organisms : LD50 (*Colinus virginianus* (Bobwhite quail)): 1,356 mg/kg
Exposure time: 14 d
Method: OECD Test Guideline 223

2-ethylhexan-1-ol:

- Toxicity to fish : LC50 (*Leuciscus idus* (Golden orfe)): 17.1 - 28.2 mg/l
Exposure time: 96 h
- Toxicity to daphnia and other aquatic invertebrates : EC50 (*Daphnia magna* (Water flea)): 39 mg/l
Exposure time: 48 h
- Toxicity to algae/aquatic plants : EC10 (*Desmodesmus subspicatus* (green algae)): 3.2 mg/l
Exposure time: 72 h
- EC50 (*Desmodesmus subspicatus* (green algae)): 11.5 mg/l
Exposure time: 72 h
- Toxicity to microorganisms : EC50 (*Anabaena flos-aquae* (cyanobacterium)): 16.6 mg/l
Exposure time: 72 h

Persistence and degradability

Components:

bifenthrin (ISO):

- Biodegradability : Result: Not readily biodegradable.
- Stability in water : Degradation half life (DT50): 2.2 d
Hydrolysis: at 60 °C
- Degradation half life (DT50): 15.6 d
Hydrolysis: at 40 °C

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

- Biodegradability : Result: Inherently biodegradable.
Biodegradation: 58.6 %
Exposure time: 28 d
Method: OECD Test Guideline 301F
Remarks: Based on data from similar materials

calcium dodecylbenzenesulphonate:

SAFETY DATA SHEET



TALSTAR 250 EC

Version	Revision Date:	SDS Number:	Date of last issue:
1.1	17.12.2025	50001563	22.12.2021
			Date of first issue: 22.12.2021

Biodegradability : Result: Readily biodegradable.
Method: OECD Test Guideline 301E

2-ethylhexan-1-ol:

Biodegradability : Result: Readily biodegradable.

Bioaccumulative potential

Components:

bifenthrin (ISO):

Bioaccumulation : Species: Lepomis macrochirus (Bluegill sunfish)
Bioconcentration factor (BCF): 1,709
Remarks: Due to the distribution coefficient n-octanol/water, accumulation in organisms is possible.

Partition coefficient: n-octanol/water : log Pow: 6.6

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Partition coefficient: n-octanol/water : log Pow: 1.99 - 18.02
Method: QSAR

calcium dodecylbenzenesulphonate:

Bioaccumulation : Species: Fish
Bioconcentration factor (BCF): 70.79
Method: QSAR

Partition coefficient: n-octanol/water : log Pow: 4.77 (25 °C)

2-ethylhexan-1-ol:

Partition coefficient: n-octanol/water : log Pow: 2.9 (25 °C)

Mobility in soil

Components:

bifenthrin (ISO):

Distribution among environmental compartments : Koc: 236610 ml/g, log Koc: 5.37
Remarks: immobile

Other adverse effects

Product:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Very toxic to aquatic life with long lasting effects.

TALSTAR 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: 22.12.2021
1.1	17.12.2025	50001563	Date of first issue: 22.12.2021

SECTION 13. DISPOSAL CONSIDERATIONS**Disposal methods**

- Waste from residues : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company.
- Contaminated packaging : Empty remaining contents.
Dispose of as unused product.
Do not re-use empty containers.
Do not burn, or use a cutting torch on, the empty drum.

SECTION 14. TRANSPORT INFORMATION**International Regulations****UNRTDG**

- UN number : UN 3352
Proper shipping name : PYRETHROID PESTICIDE, LIQUID, TOXIC (Bifenthrin)
Class : 6.1
Packing group : III
Labels : 6.1
Environmentally hazardous : yes

IATA-DGR

- UN/ID No. : UN 3352
Proper shipping name : Pyrethroid pesticide, liquid, toxic (Bifenthrin)
Class : 6.1
Packing group : III
Labels : Toxic
Packing instruction (cargo aircraft) : 663
Packing instruction (passenger aircraft) : 655

IMDG-Code

- UN number : UN 3352
Proper shipping name : PYRETHROID PESTICIDE, LIQUID, TOXIC (Bifenthrin)
Class : 6.1
Packing group : III
Labels : 6.1
EmS Code : F-A, S-A
Marine pollutant : yes

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

National Regulations

SAFETY DATA SHEET



TALSTAR 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: 22.12.2021
1.1	17.12.2025	50001563	Date of first issue: 22.12.2021

ADG

UN number	: UN 3352
Proper shipping name	: PYRETHROID PESTICIDE, LIQUID, TOXIC (Bifenthrin)
Class	: 6.1
Packing group	: III
Labels	: 6.1
Hazchem Code	: 2X
Environmentally hazardous	: yes

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

SECTION 15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Therapeutic Goods (Poisons : Schedule 6
Standard) Instrument

APVMA no.: 60987	
Prohibition/Licensing Requirements	: There is no applicable prohibition, authorisation and restricted use requirements, including for carcinogens referred to in Schedule 10 of the model WHS Act and Regulations.

The components of this product are reported in the following inventories:

TCSI	: On the inventory, or in compliance with the inventory
TSCA	: Product contains substance(s) not listed on TSCA inventory.
AIIC	: Not in compliance with the inventory
DSL	: This product contains the following components that are not on the Canadian DSL nor NDSL. 2-METHYLBIPHENYL-3-YLMETHYL (Z)-(1RS,3RS)-3-(2-CHLORO-3,3,3-TRIFLUOROPROP-1-ENYL)-2,2-DIMETHYLCYCLOPROPANECARBOXYLATE
ENCS	: Not in compliance with the inventory
ISHL	: Not in compliance with the inventory
KECI	: On the inventory, or in compliance with the inventory
PICCS	: Not in compliance with the inventory

SAFETY DATA SHEET



TALSTAR 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: 22.12.2021
1.1	17.12.2025	50001563	Date of first issue: 22.12.2021

IECSC	:	On the inventory, or in compliance with the inventory
NZIoC	:	Not in compliance with the inventory
TECI	:	Not in compliance with the inventory

SECTION 16: ANY OTHER RELEVANT INFORMATION

Revision Date : 17.12.2025

Date format : dd.mm.yyyy

Full text of other abbreviations

ACGIH : USA. ACGIH Threshold Limit Values (TLV)

ACGIH / TWA : 8-hour, time-weighted average

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

SAFETY DATA SHEET

TALSTAR 250 EC



Version	Revision Date:	SDS Number:	Date of last issue: 22.12.2021
1.1	17.12.2025	50001563	Date of first issue: 22.12.2021

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